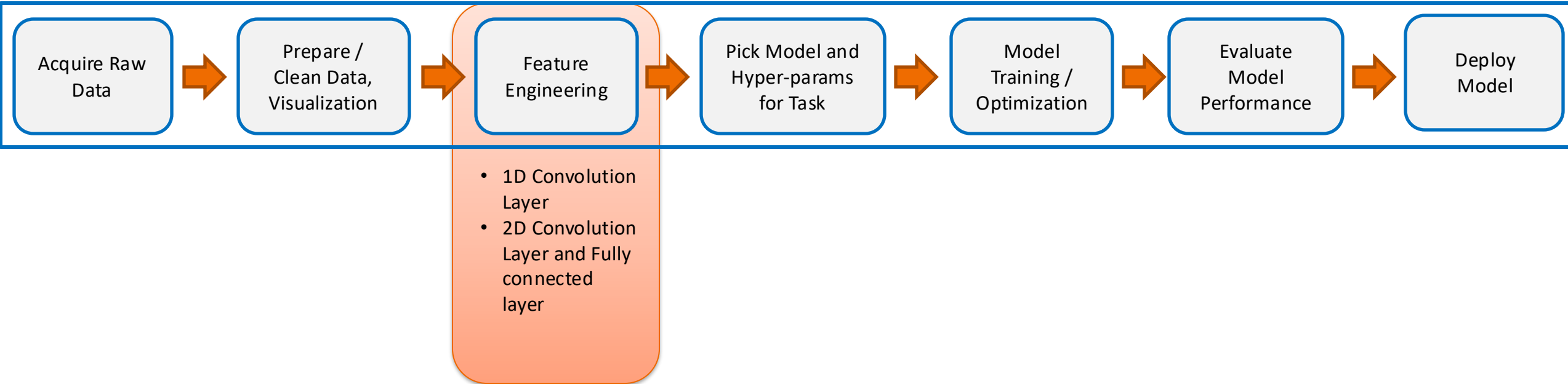


Focus for this lecture



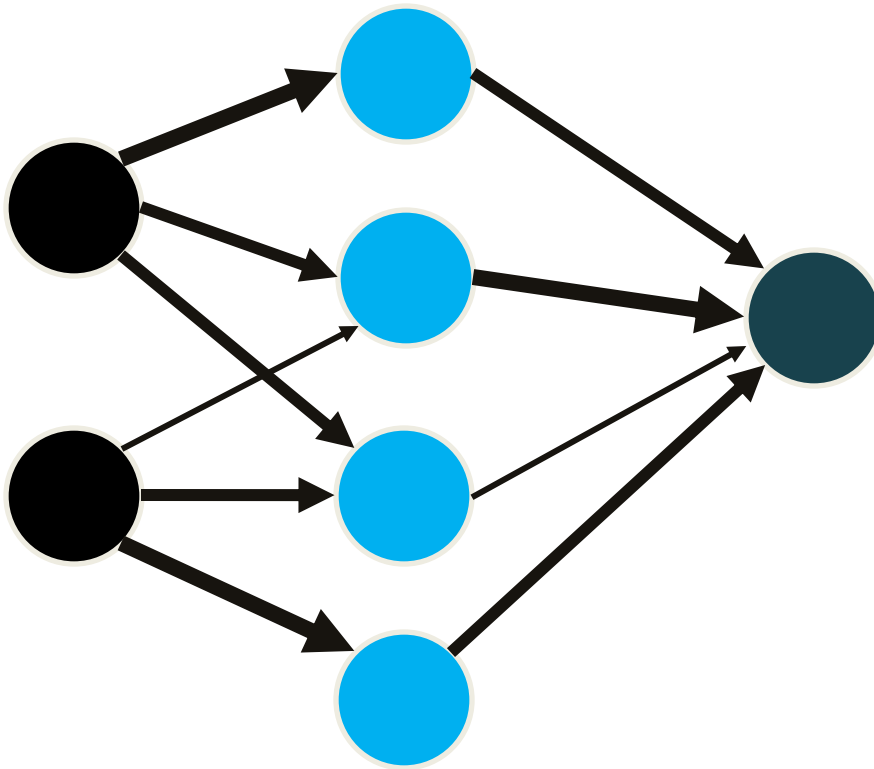
Convolution Layer

Neural Networks (Intro to CNNs)

Neural Networks

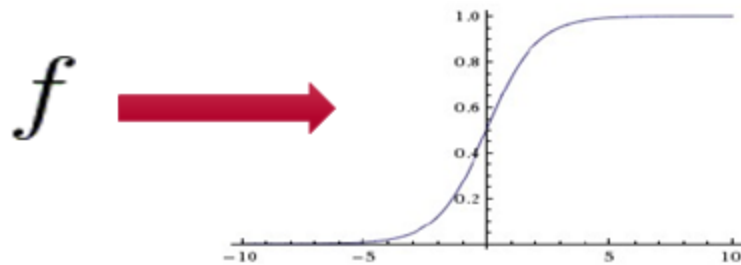
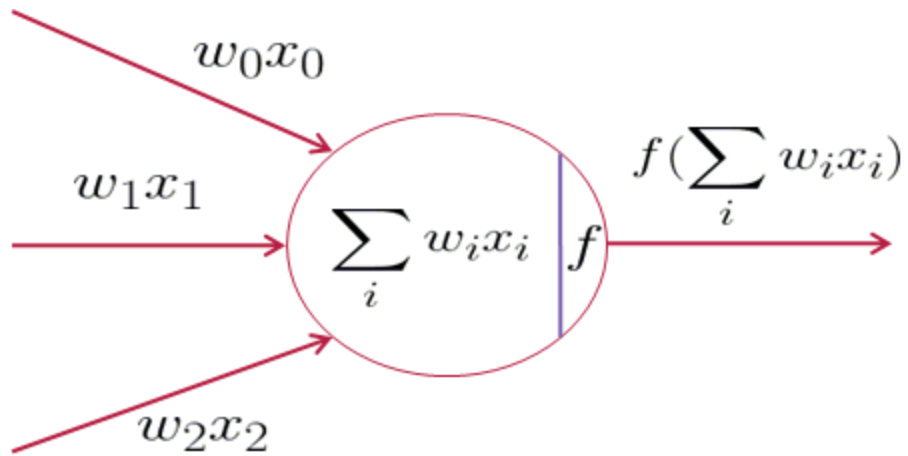
A Simple Neural Network

Input Layer *Hidden layer* *Output Layer*

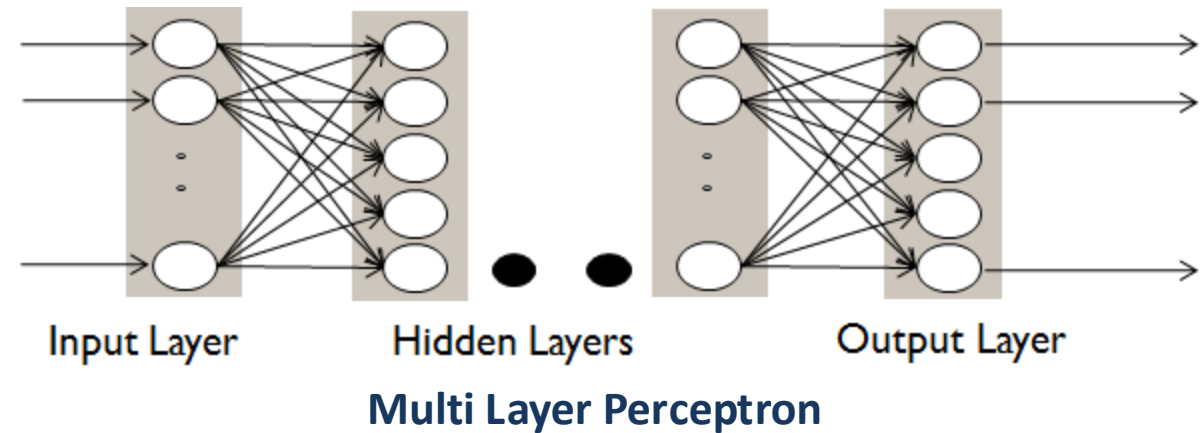
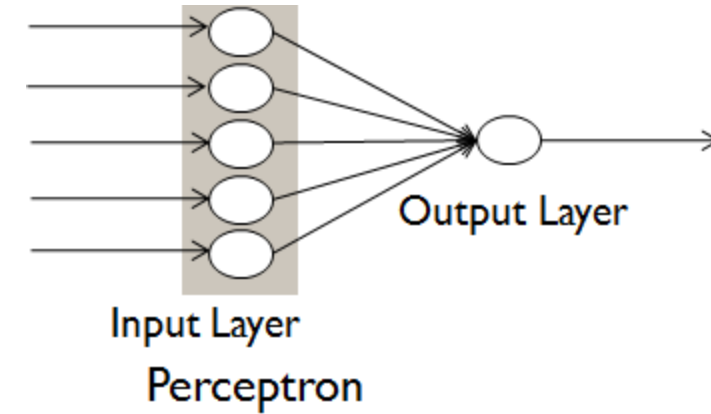


- Biologically inspired networks.
- Complex function approximation through composition of functions.
- Can learn arbitrary Nonlinear decision boundary

Neuron, Perceptron and MLP



E.g. Sigmoid Activation Function

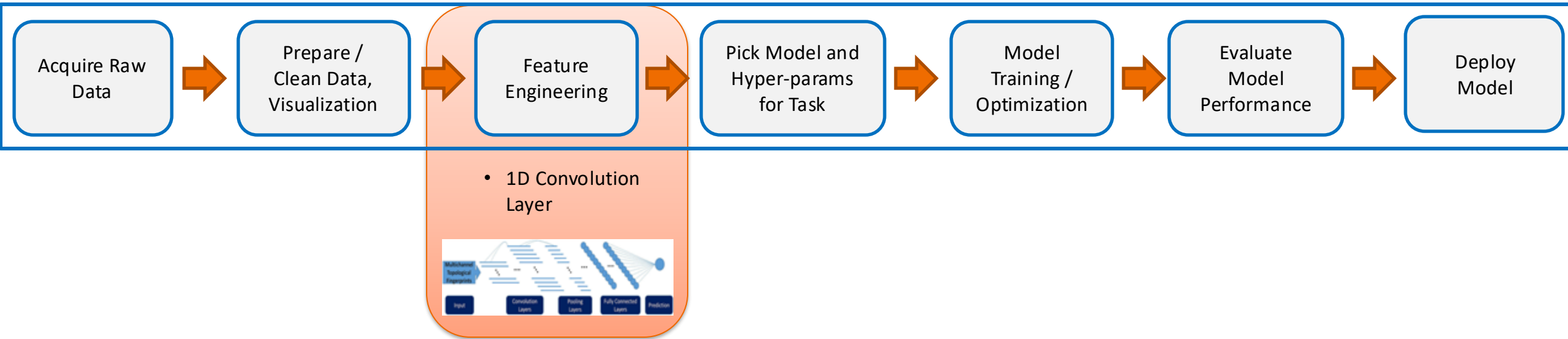


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BLANK SLIDE/Discussions

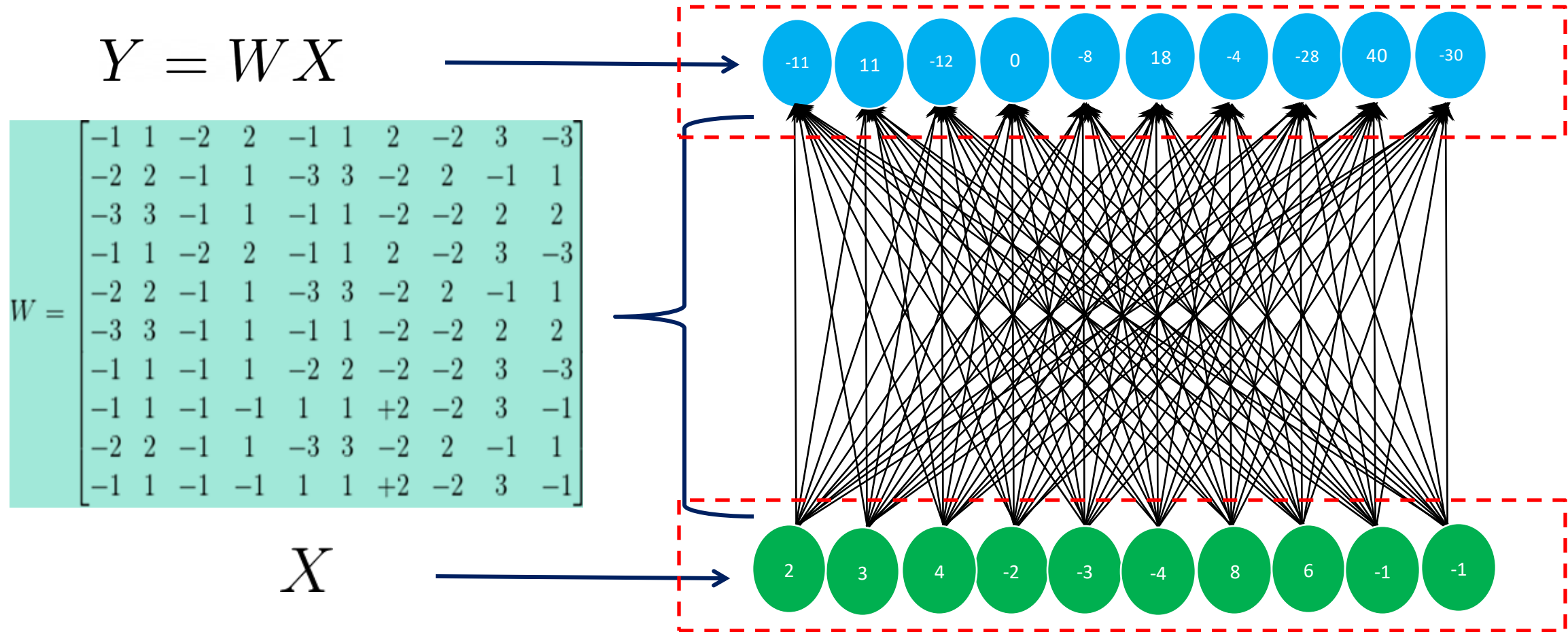
Convolutional Layer



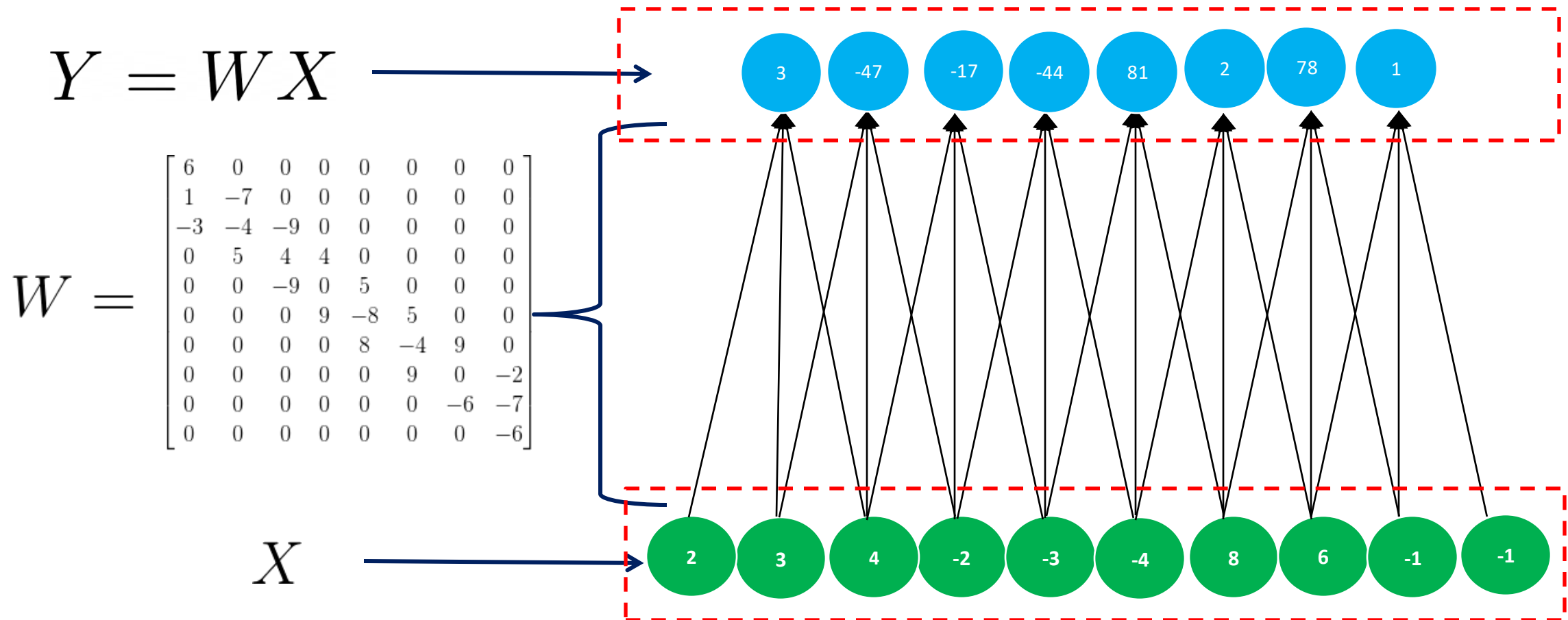


Example: 1D Convolution layer

Dense connections



What if connections are only local?



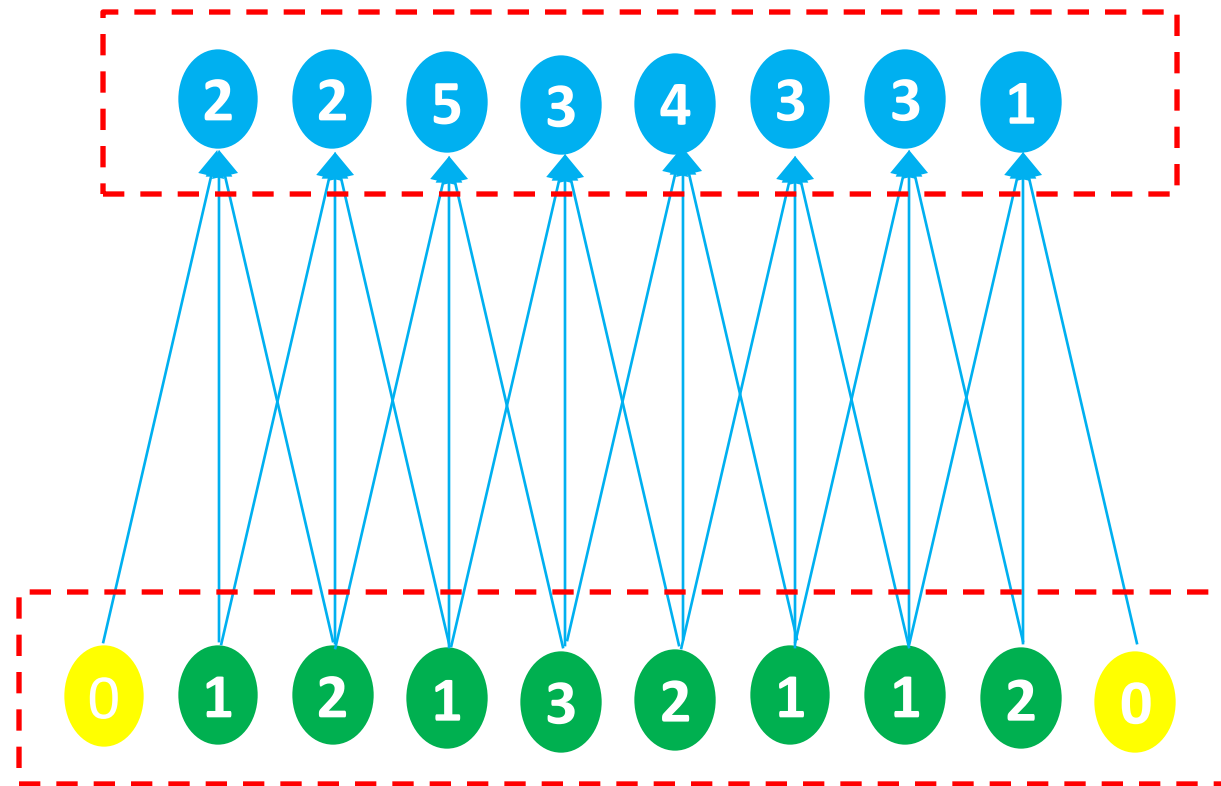
What if weights are same/shared?

Filter-1

1

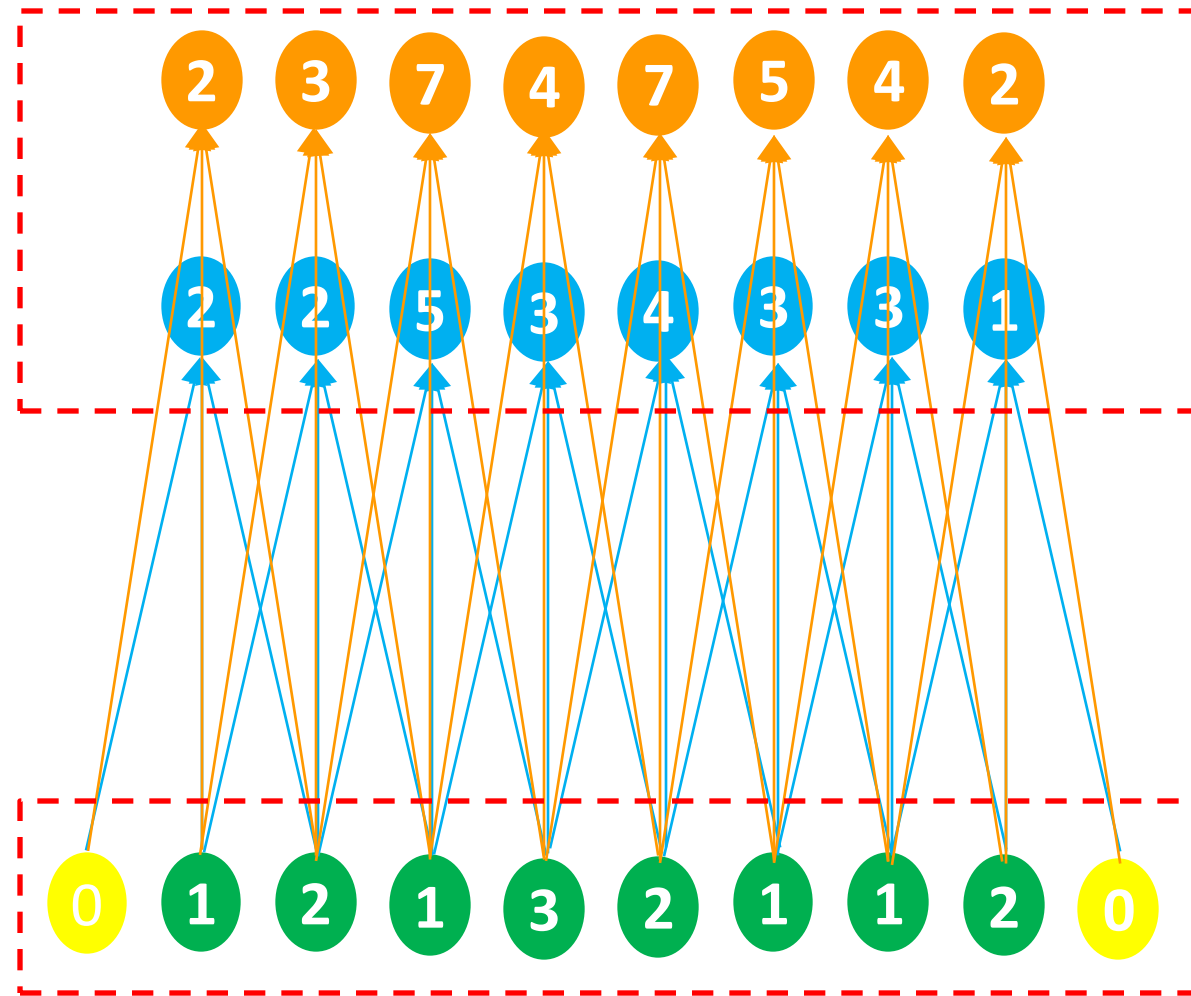
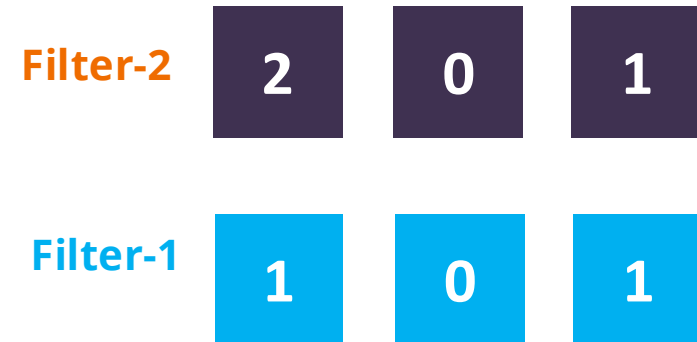
0

1



Weights that eventually we learn with
Backpropagation (filter size = #
Parameters = # weights = 3)

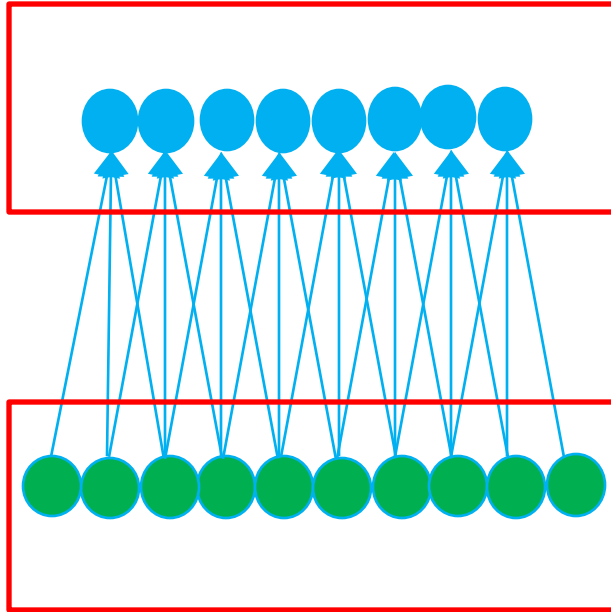
Two such filters/weights



Weights that eventually we learn with
Backpropagation (filter size = #
Parameters = # weights = 3)

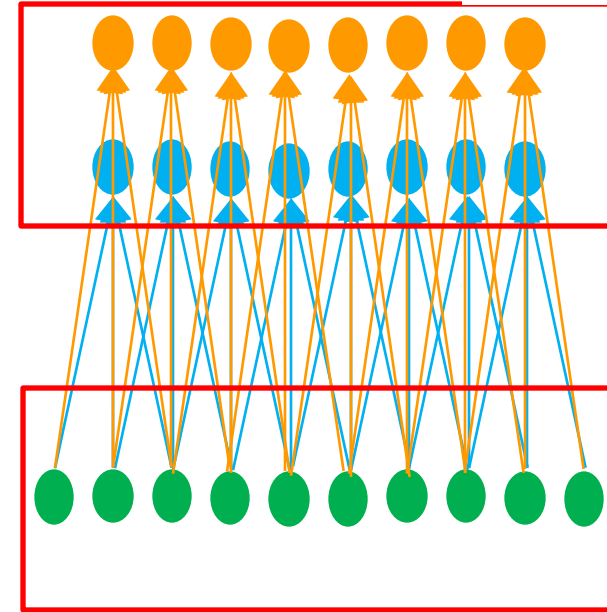
Pad Extra Samples so that
output size does not reduce

Convolution layer: Different Possibilities



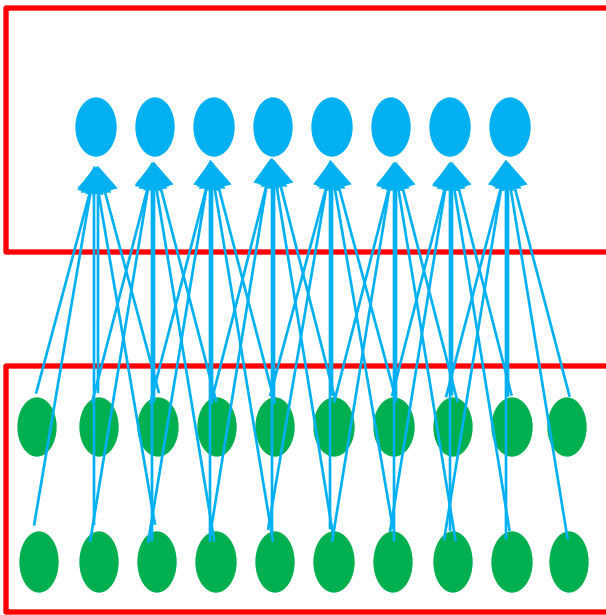
Channels:

- I/P = 1
- O/P = 1
- #Parameters = 3



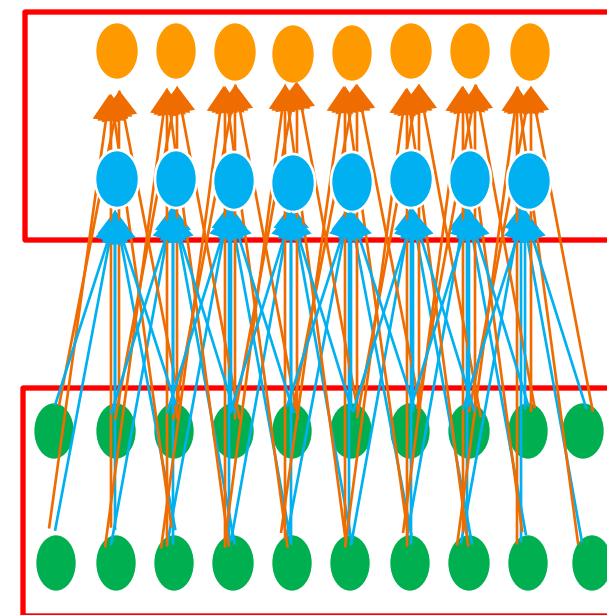
Channels:

- I/P = 1
- O/P = 2
- #Parameters = 6



Channels:

- I/P = 2
- O/P = 1
- #Parameters = 6



Channels:

- I/P = 2
- O/P = 2
- #Parameters = 12

Convolution layer: Different Possibilities



Channels:

- I/P = 1
- O/P = 1
- #Parameters = 3



Channels:

- I/P = 1
- O/P = 2
- #Parameters = 6



Channels:

- I/P = 2
- O/P = 1
- #Parameters = 6

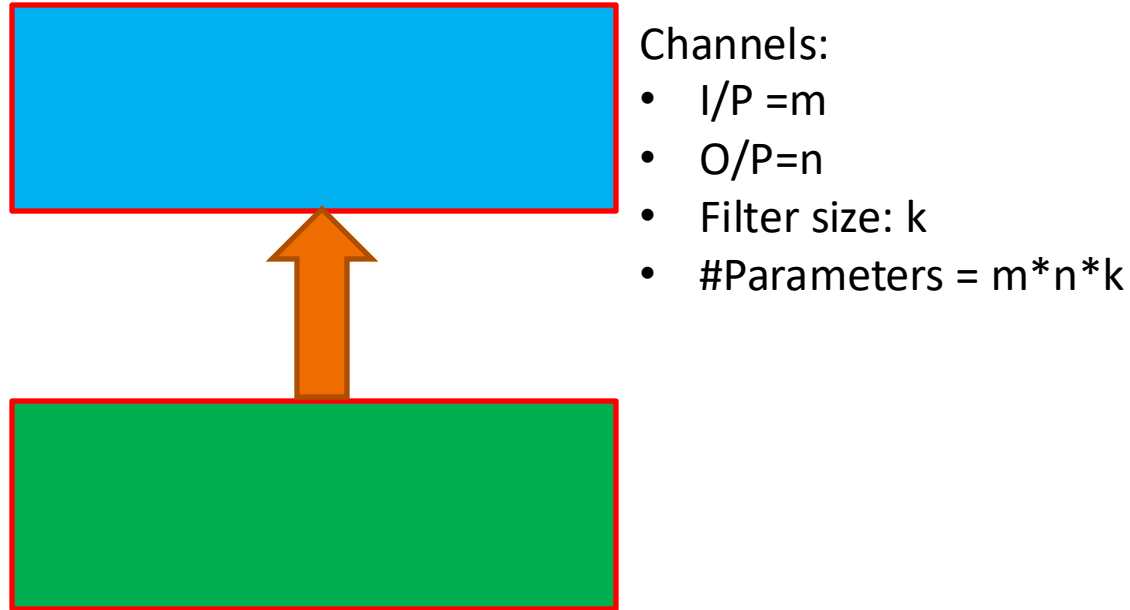


Channels:

- I/P = 2
- O/P = 2
- #Parameters = 12



We know by now ..



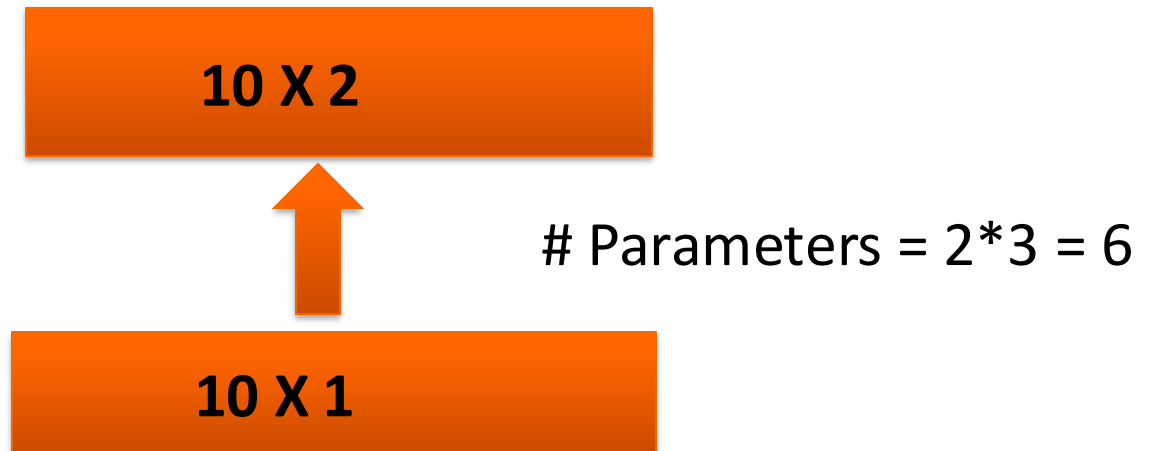
Key Words

- # Input Channels
- # Output channels
- # Weights/Parameters

Key Words

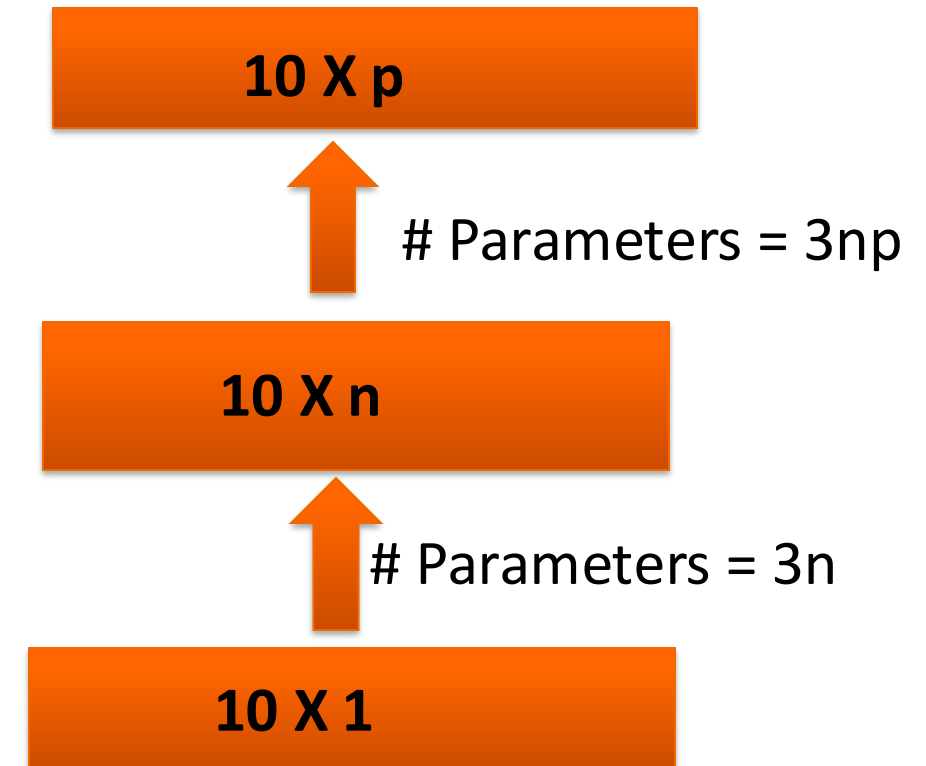
- Feature Maps/Channels
 - A representation of the data
- Filters/Weights
 - Learnable parameters (problem specific)
- Filter Size/Window Size
 - We can change. Not much to play with.
- Stride (wait)
 - Skip/reduction in size
- Padding (seen ?)
 - Extra elements so that no reduction

Pictorially Summarizing

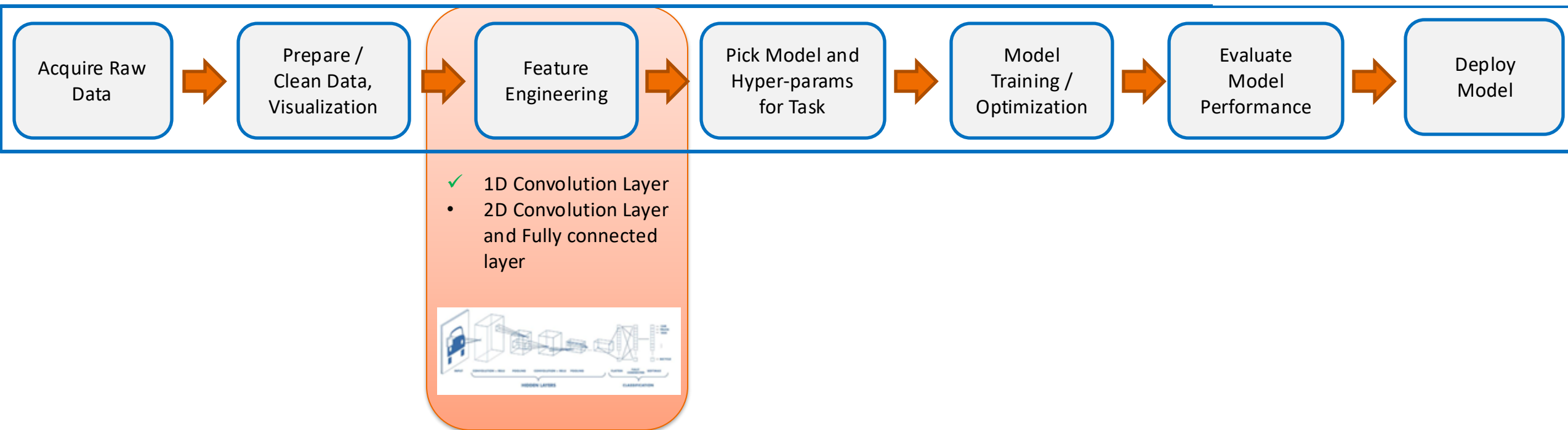


Filter size = 3

Total parameters = $3n + 3np$



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—— Convolution layer in 2D (popular) ——

Convolution

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

These are the network parameters to be learned.

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Convolution

-1	1	-1
-1	1	-1
-1	1	-1

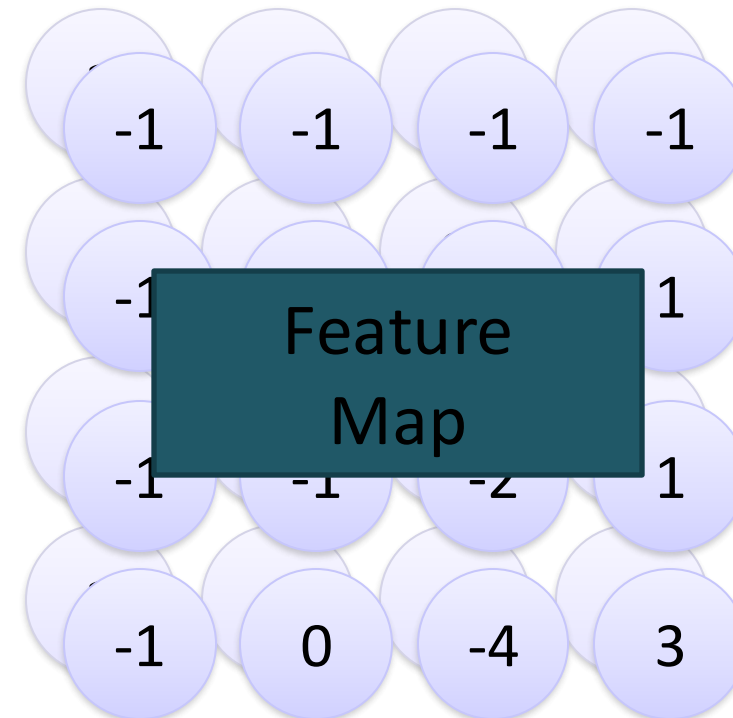
Filter 2

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

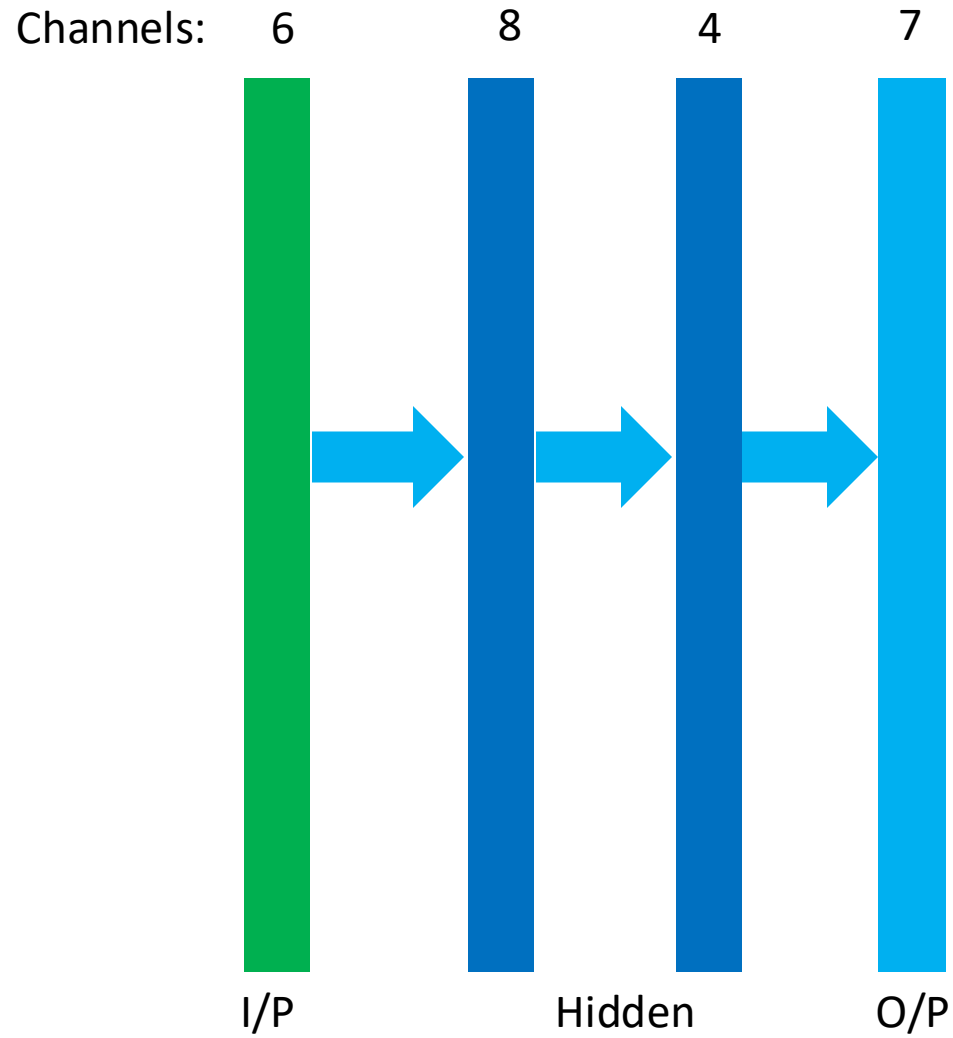
6 x 6 image

Repeat this for each filter

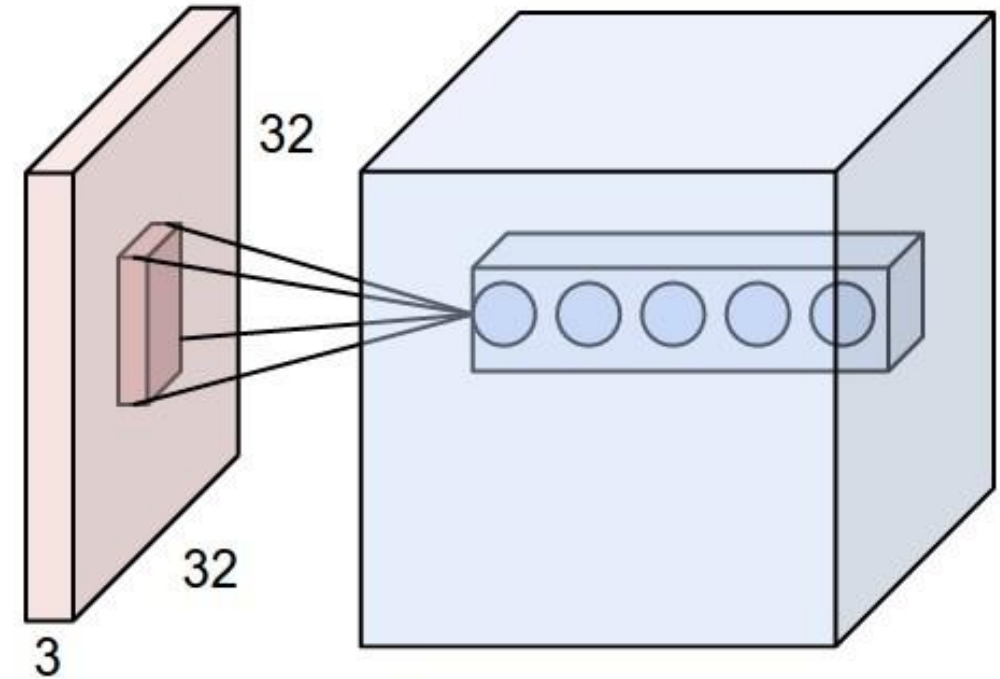


Two 4 x 4 images
Forming 2 x 4 x 4 matrix

Layer wise abstraction

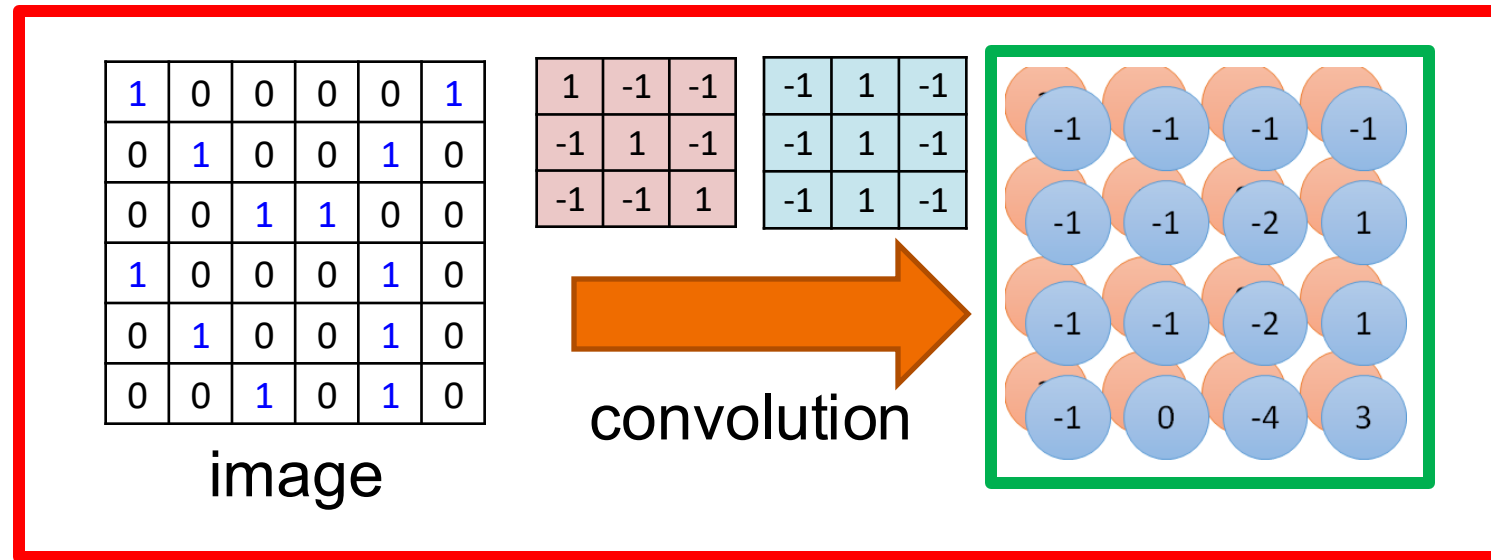


1-D Convolution

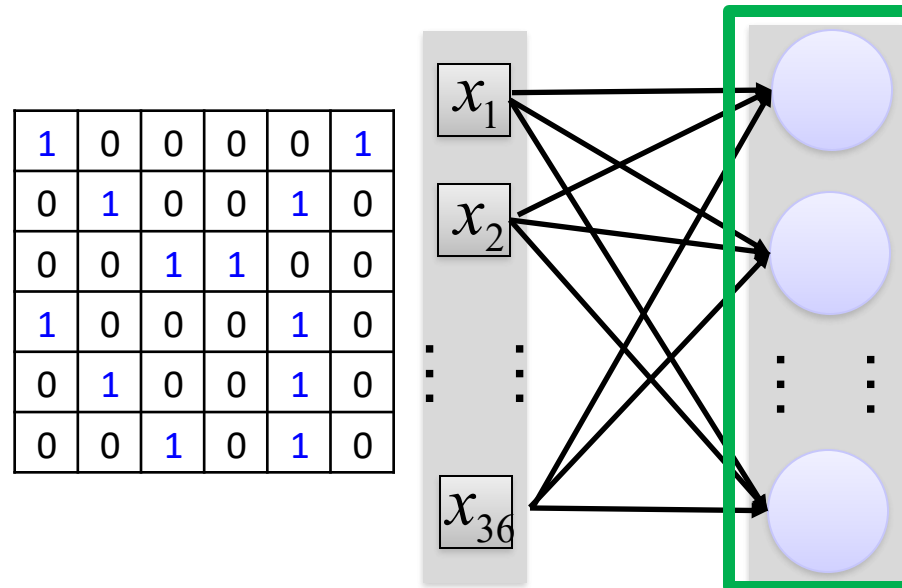


2-D Convolution

Convolution vs Fully Connected



Fully-
connected



Fully connected

- Multi layer perceptron
- Role of a classifier
- Generally used in final layers to classify the object represented in terms of discriminative parts and higher semantic entities.
- SoftMax
 - Normalizes the output.
 - K is total number of classes

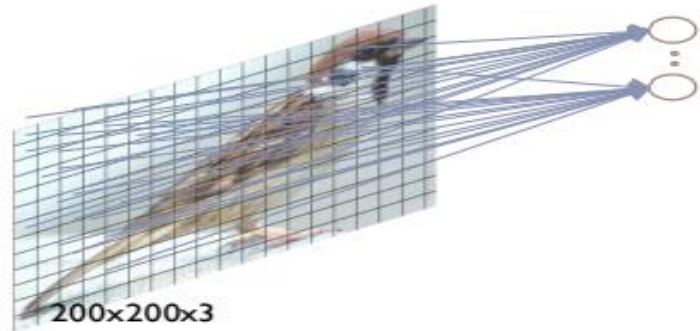
$$z_n = \frac{e^{x_n}}{\sum_{i=1}^K e^{x_i}}$$

Convolution Layer

- Maps a representation to another representation
 - Eg. A colour image to an “edge” image
 - Eg. A image to “heat maps” of parts of interest
 - Eg. “Hate words” in a sentence
 - Eg. “Location of eyes” in an image
- Role of Pattern Detector
- Sequence of convolutional layers extract higher and higher levels of patterns.

Convolution layer

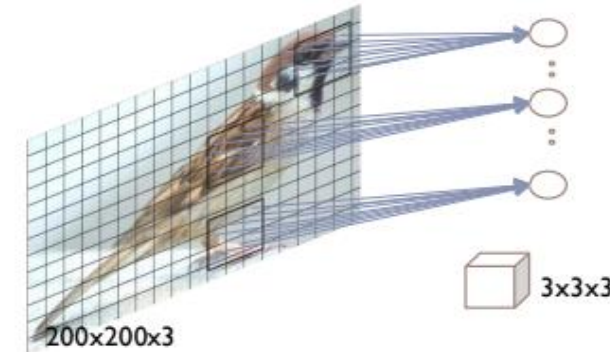
- **Fully connected layer**



- Image of size 200 X 200 and 3 colours (RGB)
- #Hidden Units: 120,000 (= 200X200X3)
- #Params: 14.4 billion (= 120K X 120K)
- Need huge training data to prevent over-fitting!

- **Locally connected layer**

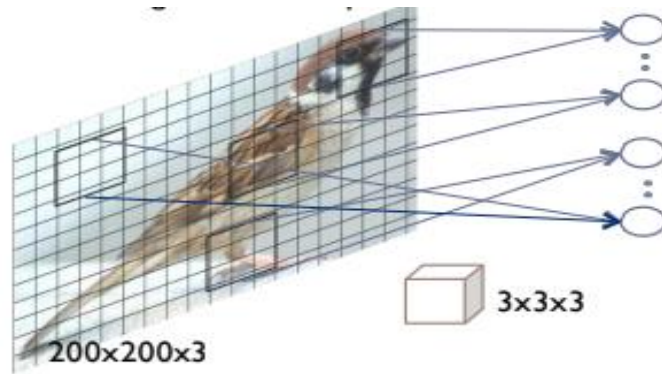
Parameter
Calculations



- #Hidden Units: 120,000
- #Params: 3.2 Million (= 120K X 27)
- Useful when the image is highly registered

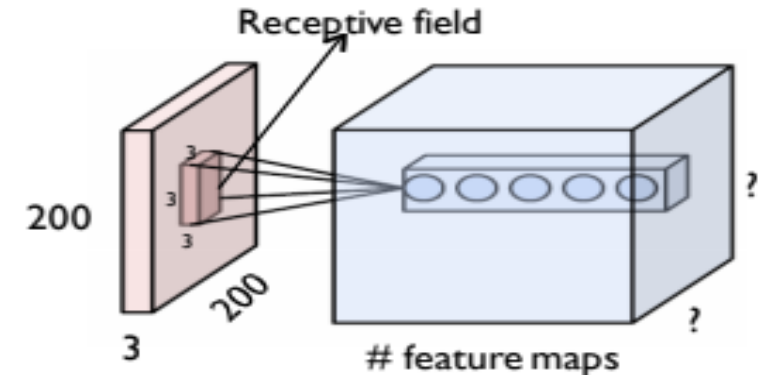
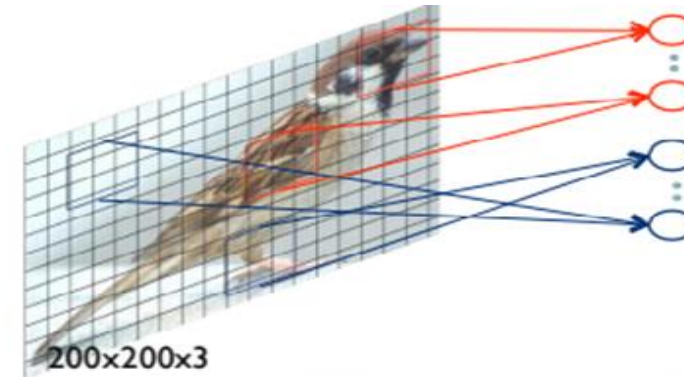
Convolution layer

- Convolutional layer with a single feature map



- #Hidden Units: 120,000
- #Params: 27 x #Feature Maps
- Sharing parameters
- Exploits the stationarity property and preserves locality of pixel dependencies

- Convolutional layer with multiple feature maps



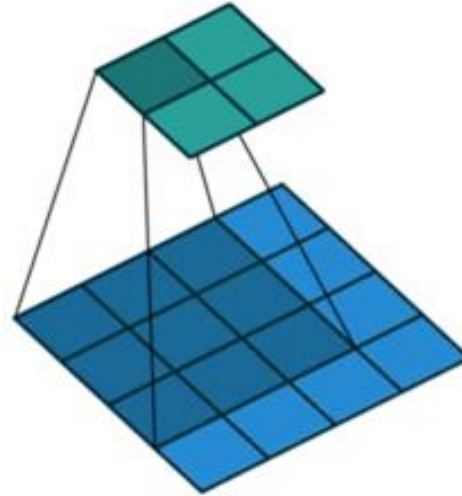
New Terms

- **1-D Convolution**
- **2-D Convolution**
- **Padding**
- **FeatureMap/Channels**
- **FeatureSize/WindowSize**
- **FilterCoeff/Weights**
- **Stride and Pool (Seen)**
- **Parameter Calculations (Seen)**

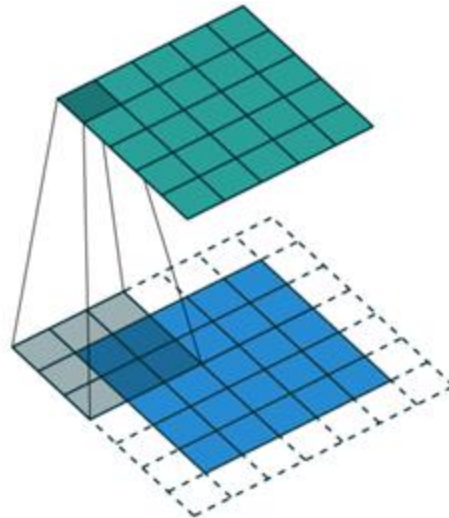
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CNNs

- Window size
- Stride
- Padding

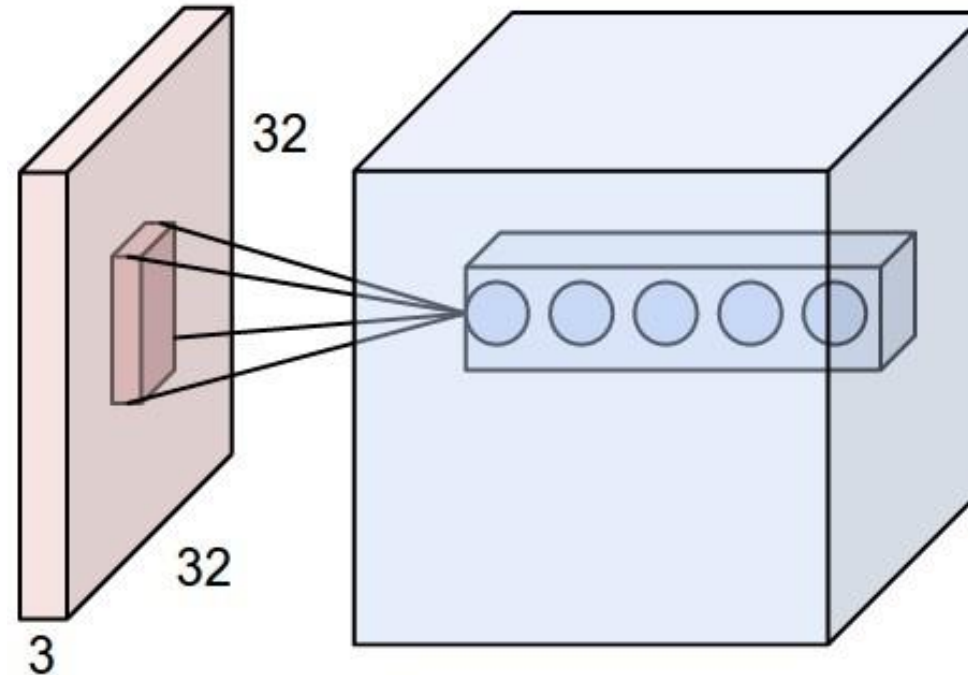


Window size: 3x3
Stride: 1
Padding: 0



Window size: 3x3
Stride: 1
Padding: 1

Input, Output Channels : Multiple Filters



DEMO: <http://cs231n.github.io/convolutional-networks/>

Input Volume (+pad 1) (7x7x3)

 $x[:, :, 0]$

0	0	0	0	0	0	0
0	0	0	0	1	1	0
0	0	1	2	2	1	0
0	2	0	2	0	2	0
0	1	2	2	1	0	0
0	2	0	1	1	1	0
0	0	0	0	0	0	0

 $x[:, :, 1]$

0	0	0	0	0	0	0
0	1	1	1	2	1	0
0	2	2	0	1	1	0
0	2	2	2	1	2	0
0	2	0	2	2	0	0
0	1	2	0	1	0	0
0	0	0	0	0	0	0

 $x[:, :, 2]$

0	0	0	0	0	0	0
0	2	2	1	0	2	0
0	2	2	1	1	1	0
0	1	2	0	1	0	0
0	1	1	2	2	0	0
0	2	0	2	1	2	0
0	0	0	0	0	0	0

Filter W0 (3x3x3)

 $w0[:, :, 0]$

0	-1	-1
1	1	0
0	0	1

 $w0[:, :, 1]$

-1	1	0
1	-1	0
1	-1	-1

 $w0[:, :, 2]$

-1	-1	0
-1	0	0
-1	1	1

Bias b0 (1x1x1)

 $b0[:, :, 0]$

1

Filter W1 (3x3x3)

 $w1[:, :, 0]$

0	-1	0
0	-1	1
0	0	0

 $w1[:, :, 1]$

0	-1	0
0	1	0
0	1	0

 $w1[:, :, 2]$

-1	-1	0
-1	-1	0
1	1	1

Bias b1 (1x1x1)

 $b1[:, :, 0]$

0

Output Volume (3x3x2)

 $o[:, :, 0]$

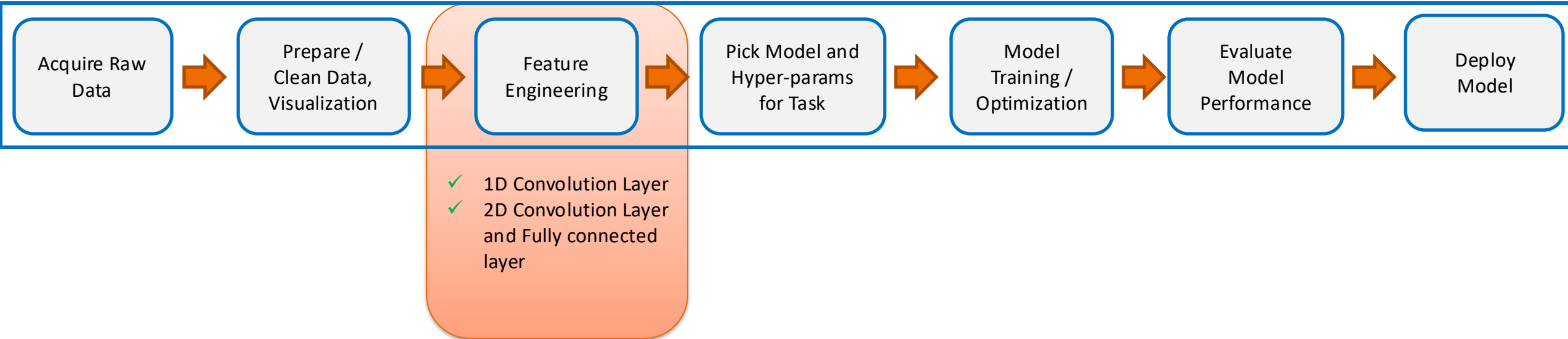
1	2	4
-2	-8	-4
-4	-4	-1

 $o[:, :, 1]$

5	3	1
-1	0	-3
-7	-9	-6

toggle movement

Summary



Thanks!!

Questions?